

Essential Information

Instructors	Dr. James Gossell (Section 901 and Section 902) Email: jegossell@alaska.edu Office: Chapman 301D Dr. Anthony Graves-McCleary (Section 903) Email: ajgravesmccleary@alaska.edu Office: Gruening 508E
Website	https://uaf-math.github.io/calcl/
Required Text	<i>OpenStax Calculus Volume 1</i> by G. Strang & E. Herman, https://openstax.org/details/books/calculus-volume-1 (optional print copy) ISBN-13: 978-1938168024
Grades	Canvas: https://canvas.alaska.edu/courses/31681
Prerequisite	MATH F151X and MATH F152X; or MATH F156X; or placement.

Description, Course Goals & Student Learning Outcomes

Calculus is one of mathematics' premiere computational tools. It has pervasive applications in all the sciences and is part of the UAF core curriculum. The two principal tools of calculus are differentiation and integration. Differentiation concerns how changes in one variable affect another. How does a population of bacteria change as time changes? How does the temperature of the ocean change as depth increases? Integration, on the other hand, is a kind of reverse process to differentiation.

Students completing the course will have the mathematical foundation to be successful in Calculus II and other courses requiring this background. Specifically, students will

- understand the role of limits in the definition of a derivative and be able to compute elementary derivatives from this definition,
- understand the definition of a continuous function and identify continuous/discontinuous functions,
- develop the skills to compute standard derivatives,
- apply derivatives to common types of applied problems,
- understand the definition of the definite integral,
- apply the Fundamental Theorem of Calculus to compute definite integrals,
- apply integration to common types of applied problems.

Class Time

There are four hours of class meetings with your primary instructor every week (MWThF). This time will be used to discuss new topics in Calculus and for a weekly quiz on Thursdays. Starting in Week 2, the last 30 minutes of the Thursday class will be used to go over the quiz so all students can leave knowing what material on that quiz has been mastered, how to work each problem correctly, and what topics need additional study.

The Tuesday Recitation (MATH F251L) will initially be spent helping students get started on first-of-semester tasks. Starting on Week 3, the Tuesday class is explicitly devoted to bolstering the underlying non-Calculus skills that are nevertheless essential to success in Calculus such as: graphing, algebra, trigonometry, exponential and logarithmic functions, and inverse functions. It will also include additional strategic homework/quiz/test prep. As a concrete example, one of the things we will do in Recitation is go over the non-Calculus math skills needed to complete the upcoming homework so that students can focus on the Calculus instead of getting bogged down in algebra or trigonometry.

Tentative Schedule

The course website contains a schedule for the semester listing the topics to be covered each class, the dates each assignment is due, the topics of every quiz, and so forth. You should consult this schedule routinely. We may make minor adjustments to the schedule, which will be announced in advance.

Office Hours and Communication

Instructors and TAs will schedule formal office hours, which will be listed on web sites accessible from the main course webpage. Any office hours in the [Student Success Center](#) are open to **all** Calculus I students regardless of who is their instructor of record. Students can also schedule meetings with their instructor outside of regular office hours.

We will use Canvas to send announcements. If we (your instructor/TA) need to contact you, we will first try to do this in class. Our second method will be to send an email to you via Canvas. Thus, you will want to make sure that the email address in Canvas is one that you check regularly. Note that in Canvas it is possible to set up text alerts. However, you must login to Canvas and adjust the setting for your account. Neither email nor text alerts are automatic.

Evaluation and Grades

Grades are determined as follows; each grade component is discussed subsequently in the syllabus.

Class Participation	7.5%
(Written) Homework	7.5%
Quizzes	20%
Midterm 1	15%
Derivative Proficiency	10%
Midterm 2	15%
Final Exam	25%
Total	100%

Letter grades will be assigned according to the following scale:

A+	97–100%	C+	77–79.9%	F	< 60%
A	93–96.9%	C	70–76.9%		
A-	90–92.9%	C-	not given		
B+	87–89.9%	D+	67–69.9%		
B	83–86.9%	D	63–66.9%		
B-	80–82.9%	D-	60–62.9%		

This scale is a guarantee; the instructors reserve the right to lower the thresholds.

Online Course Materials

All materials for this course are posted online. (See below.)

Where to find it	What you are looking for
public Calculus I webpage https://uaf-math.github.io/calc1/	syllabus and day-by-day schedule for the semester
	ALEKS logistics
	homework problem sets
	practice quizzes, proficiencies, midterms, final exams
	solutions to practice quizzes, proficiencies, midterms, final exam videos
	recitation worksheets with solutions & videos
	textbook
	in-class worksheets with solutions
Canvas	announcements and reminders
	link to Gradescope (to turn in homework)
	complete solutions to homework
	your grades
	textbook

Class Participation

Attendance and participation in class is an important part of mastering the material in Calculus (and all of your classes). Class attendance is one of the best predictors of overall course performance regardless of subject. For this reason, we want to incentivize this aspect of your education. A student is counted as having successfully participated in class if that student was in class on-time and was an active participant during the whole class. Being an active participant includes engaging in classroom tasks such as individual or group worksheets, classroom discussions, and other classroom activities. Your class participation score will be calculated as the percent of classes in which you were an active participant. Let your instructor know if you have missed class for an excused reason.

Homework

Homework assignments typically consist of a selection of problems at the end of each section of our textbook and a few additional problems. Homework is written (on paper or tablet) and turned in via Gradescope which is accessed from Canvas. Help with scanning homework can be found under [Technology Help](#) on the course webpage. Assignments are due most Mondays and Wednesdays (by 11:59 PM) in advance of the Thursday quiz. Complete worked solutions to all problems are provided in advance on Canvas. Consequently, your homework will be graded based on **effort** and **completion**. Homework can be turned in up to three days late with no penalty but will not be accepted after that unless there are extenuating circumstances. All students should earn 100% of their homework points!

The list of homework problems and homework guidelines can be found at the [Homework](#) link on the course webpage. Your first regular written homework assignment is due **Thursday August 27**.

Clearly, it is possible to short-circuit the homework by copying the solutions. It should also be clear that (a) this is a bad idea and (b) your instructor/TA will know you have done this. Our goal in providing answers and solutions is to foster the use of homework as a **learning experience**.

Quizzes

Quizzes 1 through 11 are paper quizzes and are given on Thursdays in the middle third of the class. Students will be given the opportunity to correct their quizzes in the last third of the Thursday class and can earn points back on their quiz for doing so **accurately**. The weekly quiz will usually cover the material taught in the classes held since the previous quiz; specific topics can be found in the schedule on the course website. Additionally, there will be a proctored ALEKS placement given in place of the Week 2 recitation which tests Calculus readiness. The ALEKS quiz score will be scaled out of 25 points and counted as an additional quiz grade. All other quizzes will be worth 25 points as well.

Quizzes are given under testing conditions; books, notes, and calculators are not allowed unless otherwise stated. Your performance on the quizzes **prior to corrections** is the best indicator of how well you are learning the course material and much more accurate than your homework score.

Make-up Quizzes are possible provided there is a documented, excused absence. Always contact your instructor if you miss a quiz.

Recitations

The Tuesday Recitation is taught by a teaching assistant (TA), graduate students from the Math Department who have experience teaching Calculus. Starting in Week 3, the Tuesday class is explicitly devoted to bolstering the underlying non-Calculus skills that are nevertheless essential to success in Calculus such as: graphing, algebra, trigonometry, exponential and logarithmic functions, and inverse functions. It will also include additional strategic homework/quiz/test prep. As a concrete example, one of the things we will do in Recitation is go over the non-Calculus math skills needed to complete the up-coming homework so that students can focus on the Calculus instead of getting bogged down in algebra or trigonometry.

Students may opt out of the recitations if they score 80 or higher on the proctored ALEKS assessment **and** if they have no unexcused absences during the first week. If such a student chooses to opt out of recitations, they are still allowed and encouraged to attend them, but a recitation absence will not count against them for their Class Participation portion grade. To opt out, the student must contact their instructor after Week 2 of the semester.

Midterms

There are two midterm exams this semester, to be held on Thursday, September 24 and Thursday, November 12. Note that the course webpage contains all previous Midterms (with solutions) so a student can know in advance what these are like and has lots of opportunity for practice. The midterms are the same for all sections; they are prepared and approved by all instructors teaching the course.

Make-up midterms will be given only for documented excused absences. Always contact your instructor if you miss a midterm.

Derivative Proficiency Test

A proficiency is an assessment covering a routine mechanical skill. In this course we will take a Derivative Proficiency (October 15) which will be graded on a binary scale for each problem (no partial credit). Note that the course webpage contains all previous proficiencies (with solutions) so a student can know in advance what these are like and has lots of opportunity for practice.

We have designed the grading structure in this course to prioritize and reward effort. Thus, you will have the opportunity to retake each proficiency. The highest of the scores will be used to calculate a student's grade.

Final Exam

The cumulative final exam will be held on Tuesday, December 8. Note that the course webpage contains all previous final exams (with solutions) so a student can know in advance what these are like and has lots of opportunity for practice. A make-up final exam will be given only in extenuating circumstances, for documented and excused reasons at the discretion of the instructors.

Tutoring and Resources

- The Student Success Center, located in the Rasmuson Library, offers tutors. See <https://www.uaf.edu/dms/mathlab/> for schedules and availability.
- One-on-one (or small group) tutoring is available in Chapman Building Room 210. You must schedule an appointment; see <https://www.uaf.edu/dms/mathlab/>.
- Student Support Services offers free tutoring in many subjects to students who qualify for their program.
- ASUAF offers private tutoring for a small fee (based on student income).

Rules and Policies**Incomplete Grade**

Incomplete (I) will only be given in DMS courses in cases where the student has completed the majority (normally all but the last three weeks) of a course with a grade of C or better, but for personal reasons beyond his/her control has been unable to complete the course during the regular term. Negligence or indifference are not acceptable reasons for the granting of an incomplete grade.

Late Withdrawals

A withdrawal after the deadline (currently 9 weeks into the semester) from a DMS course will normally be granted only in cases where the student is performing satisfactorily (i.e., C or better) in a course, but has exceptional reasons, beyond his/her control, for being unable to complete the course. These exceptional reasons should be detailed in writing to the instructor, department head and dean.

No Early Final Examinations

Normally, a student will not be allowed to take a final exam early. Exceptions can be made by individual instructors, but should only be allowed in exceptional circumstances and in a manner which doesn't endanger the security of the exam.

Student protections statement: The university respects and upholds the principles of due process and a fair and equitable process as specified in the Board of Regents' Policy 09.02 Student Rights and Responsibilities. For more information regarding the rights and responsibilities of students, refer to the Office of Rights, Compliance and Accountability website. You are encouraged to read the Board of Regents' policy carefully to fully understand your responsibilities to our community.

We strive to create a safe and respectful environment for all members of our community. If you have questions about expectations of you as a student or believe your rights are being violated, we encourage you to reach out to the Office of Rights, Compliance and Accountability for help. UAF reserves the right to suspend, expel or take other necessary and appropriate action in cases where a student is unable or unwilling to uphold community standards and campus safety. For more information on your rights as a student and the resources available to you to resolve problems, please go to the following site: <https://catalog.uaf.edu/academics-regulations/students-rights-responsibilities/>.

General Education statement: This course is listed as a General Education Math Course. As such this course is expected to meet the 4 general learning outcomes.

1. Build knowledge of human institutions, sociocultural processes, and the physical and natural works through the study of mathematics. Competence will be demonstrated for the foundational information in each subject area, its context and significance, and the methods used in advancing each.
2. Develop intellectual and practical skills across the curriculum, including inquiry and analysis, critical and creative thinking, problem solving, written and oral communication, information literacy, technological competence, and collaborative learning. Proficiency will be demonstrated across the curriculum through critical analysis of proffered information, well-reasoned solutions to problems or inferences drawn from evidence, effective written and oral communication, and satisfactory outcomes of group projects.
3. Acquire tools for effective civic engagement in local through global contexts, including ethical reasoning, intercultural competence, and knowledge of Alaska and Alaska issues. Facility will be demonstrated through analyses of issues including dimensions of ethics, human and cultural diversity, conflicts and interdependencies, globalization, and sustainability.
4. Integrate and apply learning, including synthesis and advanced accomplishment across general and specialized studies, adapting them to new settings, questions and responsibilities, and forming a foundation for lifelong learning. Preparation will be demonstrated through production of a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation and reflection.

Student Academic Support:

- Communication Center (907-474-7007, uaf-commcenter@alaska.edu, Student Success Center, 6th Floor Room 677 Rasmuson Library)
- Writing Center (907-474-5314, uaf-writing-center@alaska.edu, Student Success Center, 6th Floor Room 677 Rasmuson Library)

- UAF Math Services (907-474-7332, uaf-traccloud@alaska.edu)
 - Drop-in tutoring, Student Success Center, 6th Floor Room 672 Rasmuson Library)
 - 1:1 tutoring (by appointment only), 6th Floor Room 677 Rasmuson Library
 - Online tutoring (by appointment only) available
 - <https://www.uaf.edu/dms/mathlab/>, available at the Student Success Center
- Developmental Math Lab (Gruening 406, <https://www.uaf.edu/deved/math/>)
- The Debbie Moses Learning Center at CTC (907-455-2860, 604 Barnette St, Room 102, <https://www.ctc.uaf.edu/student-services/student-success-center/>)
- For more information and resources, please see the Academic Advising Resource List <https://www.uaf.edu/advising/students/index.php>

Student Resources:

- Disability Services (907-474-5655, uaf-disability-services@alaska.edu, 110 Eielson Building)
- Student Health & Counseling [free counseling sessions available] (907-474-7043, <https://www.uaf.edu/chc/appointments.php>, Whitaker Building, Room 206,
- Health, Safety & Security Bldg - same building as Fire and Police)
- Office of Rights, Compliance and Accountability (907-474-7300, uaf-orca@alaska.edu, 3rd Floor, Constitution Hall)
- Associated Students of the University of Alaska Fairbanks (ASUAF) or ASUAF Student Government (907-474-7355, asuaf.office@alaska.edu, Wood Center 119)

Disability services statement: UAF is committed to creating a learning environment that meets the needs of its diverse student body. If you have a disability, or think you might have a disability, please contact Disability Services (DS) to request a reasonable accommodation. Accommodations are not retroactive, so students should request accommodations as early as possible, since they might take time to implement. Students should notify DS at any time during the semester if adjustments to their communicated accommodation plan are needed, and must request their accommodations each semester. More information is available online at <https://www.uaf.edu/disability>, and students can reach DS at uaf-disability-services@alaska.edu or 907-474-5655.

Academic Dishonesty: Academic dishonesty, including cheating and plagiarism, will not be tolerated. It is a violation of the Student Code of Conduct and will be punished according to UAF procedures.

ASUAF advocacy statement: The Associated Students of the University of Alaska Fairbanks (ASUAF), UAF's student government, works to represent student voices and improve the student experience. Students are encouraged to get involved in shaping policies, sharing feedback, and helping address issues that impact campus life. ASUAF also offers support in navigating university processes and connecting with the right resources when challenges arise. To get involved or seek support, visit the ASUAF office or email asuaf.office@alaska.edu.

Nondiscrimination statement: The University of Alaska is an equal opportunity/equal access employer, educational institution and provider. The University of Alaska does not discriminate on the basis of race, religion, color, national origin, citizenship, age, sex, physical or mental disability, status as a protected veteran, marital status, changes in marital status, pregnancy, childbirth or related medical conditions, parenthood, sexual orientation, gender identity, political affiliation or belief, genetic information, or other legally protected status. The University's commitment to nondiscrimination, including against sex discrimination, applies to students, employees, and applicants for admission and employment. Contact information, applicable laws, and complaint procedures are included on UA's statement of nondiscrimination available at www.alaska.edu/nondiscrimination.

For more information, contact:

UAF Office of Rights, Compliance and Accountability
1692 Tok Lane
3rd floor, Constitution Hall, Fairbanks, AK 99775
907-474-7300
uaf-orca@alaska.edu